

Haonan Bai

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[GitHub](#) | [LinkedIn](#)

EDUCATION

Master of Applied Science (Thesis) - Electrical and Computer Engineering

2026 - Present

University of British Columbia

Vancouver, Canada

Bachelor of Engineering - Artificial Intelligence: Systems and Technologies

2021 - 2026

The Chinese University of Hong Kong

Hong Kong

RESEARCH EXPERIENCE

Data Poisoning on Reasoning Large Language Models

Jun 2025 - Dec 2025

University of California, Santa Barbara

Advisor: [Prof. Xin Eric Wang](#)

- Proposed *ItTakesTwo*, a two-stage paradigm that improves the scalability and stealthiness of data poisoning attacks against large reasoning models (LRMs), while preserving model utility.
- Evaluated the attack success rate and auditability across various LRMs.

BiasPainter: Fairness Analysis of AI Image Generators

May 2023 - Sep 2023

The Chinese University of Hong Kong

Advisor: [Prof. Michael R. Lyu](#)

- Proposed a metamorphic testing framework that can accurately, automatically and comprehensively trigger social bias in image generation models.
- Designed and built the image dataset, prompt dataset, and metamorphic relations for bias evaluation experiments.
- Performed extensive evaluations on six widely deployed commercial and research image generation models.

Bias Evaluation on Open-Domain Question Answering Systems

Sep 2022 - Feb 2023

The Chinese University of Hong Kong

Advisor: [Prof. Wenxuan Wang](#)

- Conducted experiments on triggering bias in question answering (QA) systems.
- Designed an algorithm to evaluate the bias of different QA models.

PUBLICATIONS

- Wenxuan Wang, **Haonan Bai**, Jen-tse Huang, Yuxuan Wan, Youliang Yuan, Haoyi Qiu, Nanyun Peng, Michael R. Lyu. "New Job, New Gender? Measuring the Social Bias in Image Generation Models", *ACMMM* 2024.
- Yuxuan Wan, Wenxuan Wang, Pinjia He, Jiazhen Gu, **Haonan Bai**, Michael Lyu. "BiasAsker: Measuring the Bias in Conversational AI System", *ESEC/FSE* 2023.

HONOURS & AWARDS

First-Class Honours

2026

The Chinese University of Hong Kong

Highest degree classification

ELITE Stream Qualification

2026

Faculty of Engineering, The Chinese University of Hong Kong

Advanced Engineering Program

Oral Presentation

2024

The 32nd ACM International Conference on Multimedia

Top 4% among all submissions

Undergraduate Research Summer Internship Best Project Award

2023

Faculty of Engineering, The Chinese University of Hong Kong

Top 10% among all participants

FLY Award

2022

Chung Chi College, The Chinese University of Hong Kong

Funded summer study at UC Berkeley

CUHK Scholarship for Remarkable Endeavour for Admission

2021

The Chinese University of Hong Kong

Recognized perseverance in pursuing higher education

WORK EXPERIENCE

Data Scientist Intern

Dec 2023 - Jan 2024

SmartAge Intelligence Limited

- Designed and implemented Python scripts to retrieve motion data from IoT devices and to update the database regularly.
- Designed algorithms to analyze motion data and provide alerts when abnormal motion patterns were detected

Student Helper

Mar 2023 - Jun 2023

CUHK Business School

- Designed and implemented Python web crawlers to collect PDFs from databases.

SKILLS

Programming Languages Python | C/C++ | Java | JavaScript

Tools & Libraries PyTorch, NumPy, Pandas, OpenCV | CUDA, OpenMP, MPI | React, Node.js